

Automatic Path Planning for Navigated Pedicle Screw Surgery Based on Deep Neural Network*

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Abstract— In this paper, we aim to achieve intelligently automatic planning for navigated pedicle screw surgery. To this end, deep neural network is introduced into the procedure of path planning. First, the surgical path (line) is represented as two control points: the entry point and the direction indication point. Then, features on CT image of spine are extracted utilizing a designed multi-layer 3D convolutional neural network, a 3D U-Net-like segmentation network is designed to rebuild vertebral body (VB) of spine and a fully connection network is connected to learn two control points on the segmented VB. Extensive experimental results show that our network can perform a pixel-wise multi-class segmentation and quickly locate the landmarks on surgical path in the navigated spine surgery with accepted accuracy. This work provides a reference for robot-assisted surgical path planning.

I. INTRODUCTION

Surgical path planning is a precondition and a key step in image-guided robot-assisted orthopedic surgery^[1]. Traditional methods for surgical path planning mostly rely on the experience of surgeons, other factors such as surgeon's physiological state and subjective judgment may also affect the quality of path planning, which means that the stability and efficiency of surgery cannot be effectively guaranteed. Therefore, more efficient and intelligent surgical path planning techniques should be considered for image-guided surgery.

Machine learning especially the deep learning, has been widely used in image segmentation and feature recognition^{[2][3]}. Deep learning can replace the traditional complicated and time-consuming human-computer interaction without dramatically decreasing the segmentation and recognition accuracy^[4]. In the field of orthopedics, Hao Chen et al.^[5] extracted high-dimensional features through deep convolutional neural network (CNN), and achieved automatic identification and positioning of spinal vertebra on three-dimensional (3D) CT images. BoostNet designed by

Shuo Li et al.^[6] integrates statistical method into the feature extraction process of CNN, and can automatically adapts to different feature's type during identification. The MSE (mean square error) of identification results on idiopathic scoliosis images reaches 0.00068 (validation image set) and 0.0046 (test image set). Rens Janssens et al.^[7] proposed SpineNet network based on fully CNN for spinal segmentation and the accuracy of SpineNet is higher than traditional optimal segmentation methods. Shouhei Hanaoka et al.^[8] adopted L-PDM (landmark point distribution model) probability model to locate five key landmarks on the spine, and selected the hypothesis with largest posterior probability as the anatomical testing result, achieving the detection of abnormal vertebral bodies (VBs) in body's CT images. Amin Suzani et al.^[9] used deep learning to locate the center point of a VB of lumbar spine and improved the result by using maximum component center point of the threshold value, thus achieving automatic positioning and recognition of the center point. Dong Yang et al.^[10] proposed an automatic location method for the initial geometric landmarks on femoral surface in 3D MR images. Based on the results of CNN classifier, anatomical feature landmarks were located on the femur according to geometric clues of the surface mesh.

Current researches on deep learning in clinical medicine mainly focus on anatomical and pathological feature extraction from diagnosis images of human bones from medical images, which are mostly used for surgical diagnosis and rarely directly used for surgical intervention, especially for automatic planning in spinal surgical paths. We believe that there are two reasons for this phenomenon: first, compared with a large number of image analysis with deep learning, application of deep learning in surgical intervention is relatively small; with the intelligentization trend of surgical intervention, role of artificial intelligence will become more and more important for image-guided surgery. Secondly, traditional planning method of surgical path relies on the experience of surgeons, and diversity exists in the expression of the path, which may lead to a large uncertainty in the learning process of deep network.

Navigated pedicle screw implanting is a typical case of spinal surgery. In this study, deep neural network is adopted for surgical path planning and the automatic learning method in surgical path was studied with the spinal pedicle screw surgery as the research object.

II. METHODS

Automatic planning method of spinal pedicle screw surgical route is proposed, which includes the representation of surgical path, and the designs of segmentation network and location network.

*Research supported by the Beijing Natural Science Foundation (Grant no. Z170001), and the National Key R&D Program of China (Grant no. 2018YFB1307603).

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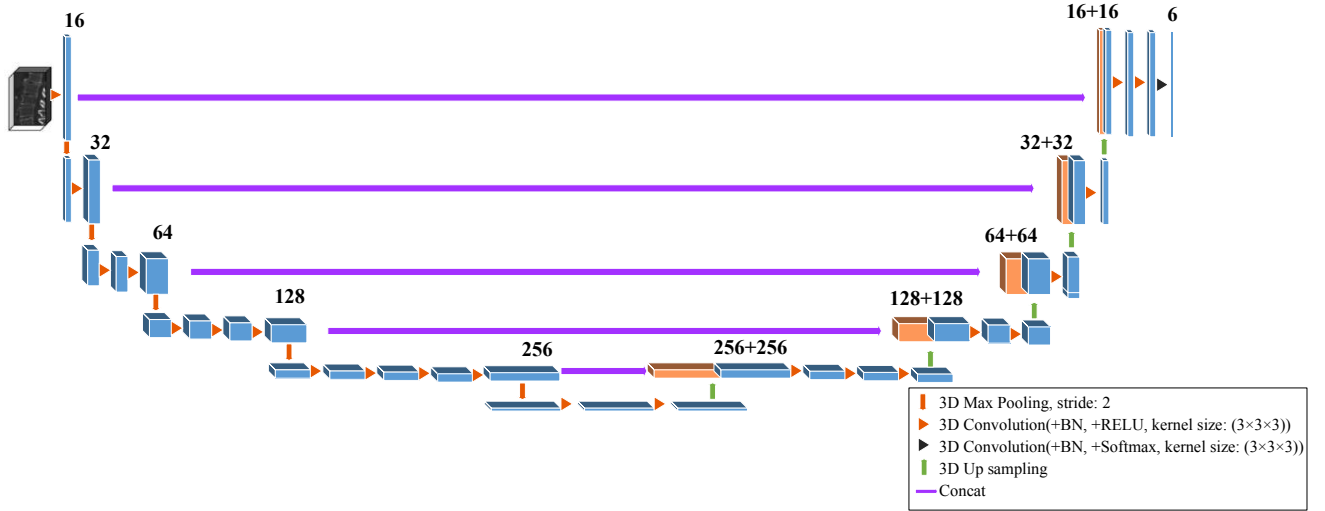


Figure 1. Architecture of the U-Net-like segmentation network. The number above each block indicates the num of channels.

A. Path Representation for Navigated Pedicle Screw Surgery

Path of pedicle screw surgery can be represented as a straight line segment on the CT image of spine. For the purpose of automatic learning, such pathways can be abstracted into two key points: one is the surgical entry point that locates on the spine surface for planning the location of pedicle screw implantation; the other is the surgical path orientation point, locates inside the spine for constraining the direction of pedicle screw. From the perspective of surgical safety, this research sets the control point at the thinnest part of the pedicle wall. Information provided by two control points indicates the path of pedicle screw implantation in the spine.

B. 3D U-Net-like Segmentation Network

In this research, 3D spinal CT images are used to segment VBs. Deep learning has been widely used in 2D image segmentation, and the fully convolutional network (FCN) [11] is the first neural network successfully used for image segmentation, which replaces fully connected network with the convolutional network so that the network can accept any size of image input and at the same time generate segmentation diagram of the original image. However, the fully convolutional network generates images directly from the extracted high-dimensional features and the generated segmentation graph is very rough. In order to solve the above problem, some people put forward a pyramid structure [12], in which input images are put into different proportions and applied to the fully convolution network respectively, the predicted results are fused to obtain the final output. This method needs to be retrained each time for input images of different sizes, which requires a large amount of computation and is difficult to be used in 3D images. Others put forward U-Net [13] network, which can restore spatial resolution by applying multi-scale features of coding stage to decoding stage by establishing *encoder-decoder* structure, and realize sharing of image features through cross-layer connection, so as to better collect image features. In order to better process 3D image information, a 3D U-Net [14] like network is adopted here for our purpose. Fig. 1 shows a schematic drawing of the

architecture of the employed segmentation network, which consists of two parts, i.e., the encoder part and the decoder part. The encoder part focuses on analysis and feature representation learning from the input data while the decoder part generates segmentation results, relying on the learned features from the encoder part. Shortcut connections are established between layers of equal resolution in the encoder and decoder paths. Unlike traditional 3D U-Net networks, our network is deeper and wider: In the encoder part, the size of the input image is halved by maximum pooling layer. The whole encoder part will go through the maximum pooling layer for 6 times and the input $128 \times 128 \times 128$ image will generate a feature map of $4 \times 4 \times 4$ size after passing through the encoder part. The decoder part uses 6 up-sampling to recover the extracted feature graph to $128 \times 128 \times 128$ size. After each up-sampling, the features of the corresponding coding layer and decoding layer are splicing together by cross-layer connection to increase the network's ability to recognize features of different sizes. Softmax layer was used as the last layer to classify the images generated by the decoder according to each individual pixel, and finally generate the segmentation diagram with channel number of 6 and size of $128 \times 128 \times 128$, corresponding to five VBs and background areas respectively.

Previous studies show that small convolutional kernels are more beneficial for network training [15]. For our segmentation network, most convolutional layers use kernel size of $3 \times 3 \times 3$ (some convolutional layers use kernel size of $5 \times 5 \times 5$ to extract larger feature maps) and strides of 1 and all max pooling layers uses kernel size of $2 \times 2 \times 2$ and strides of 2. In the convolutional and deconvolutional blocks of our network, Batch normalization (BN) [16] and rectified linear unit (ReLU) [17] are adopted.

Training process: Training of the segmentation network is done in two steps. First, the segmentation network is trained for binary segmentation of the lumbar spine, where all vertebrae have the same label. The segmentation network has to distinguish between spine and background. After that, it is

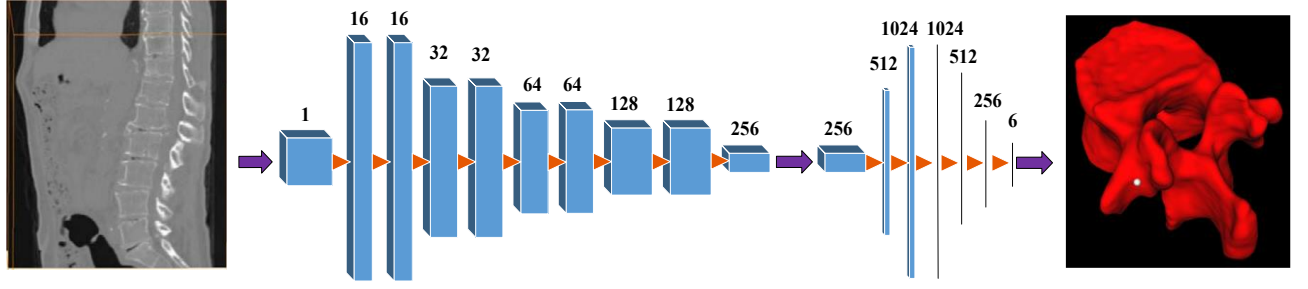


Figure 2. Schematic view of the location network.

trained for multi-class segmentation where one class corresponds to one vertebra. Pertained weights of the binary segmentation network are used to initialize the multi-class segmentation.

We use *DSC* (dice similarity coefficient) [18] as the loss function of the network in training process. *DSC* can be used to measure the degree of similarity of two contour regions, which is defined as follows:

$$DSC(A, B) = 2 | A \cap B | / (| A | + | B |) \quad (1)$$

where *A* and *B* represent the set of points contained in the two contour areas.

We took the actual segmentation graph and the network output segmentation graph as the input of *DSC*, and used stochastic gradient descent (SGD) optimizer to optimize the loss. The learning rate of the network was set at 0.001, and the learning rate was set to decay every 100 epochs, which enabled the network to converge to a better feature space.

Considering there is an imbalance between background voxels and vertebra voxels (the background class has more voxels than other classes), we reduce the weights of the background voxels and increases the weights of the vertebra voxels to balance the influence [19].

Testing process: Spinal CT images of $128 \times 128 \times 128$ are input into the segmentation network after normalization to output the segmentation map of the spine. The network can quickly and directly segment the five VBs of the spine and reconstruct the VBs based on the segmentation map.

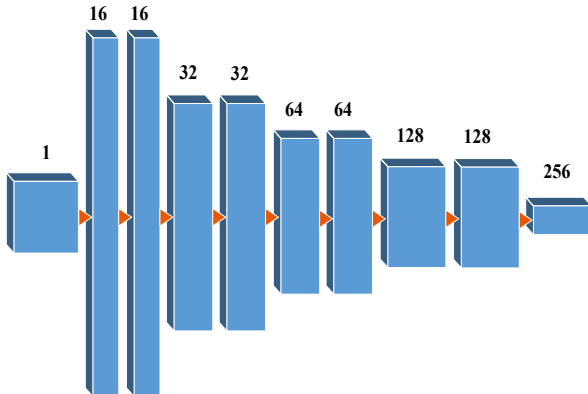


Figure 3. Architecture of the feature extraction network

C. Localization network

Segmentation network allows us to segment each individual lumbar vertebra from the CT data and reconstruct the spine. The goal of this stage is then to locate surgical path points. Inspired by previous work [20][21], we also formulate the path points localization as a multi-variate regression problem. We designed a 3D deep learning network (Fig. 2). The network consists of feature extraction network and control point location network.

Feature extraction network (Fig.3) adopts a 3D fully convolution structure, and the whole network is divided into several convolution modules. Each convolution module is composed of two convolution layers and one maximum pooling layer. The convolutional layers have a kernel size of $3 \times 3 \times 3$ and use *valid* padding so that the output size of the convolutional layer is the same as the input size. The max pooling has a stride of two and halves the spatial dimensionality. The input to the localization network is a 3D image of size $128 \times 128 \times 128$. After the five repetitions, the size of the feature maps is $4 \times 4 \times 4$.

The input of the control point location network (Fig. 4) is the spinal feature map extracted by feature extraction network. It is resized by a convolutional layer with a kernel size of $4 \times 4 \times 4$ to the size of $1 \times 1 \times 1$ with 512 features. This process is followed by two $1 \times 1 \times 1$ convolutions which upgrade the dimension of the image so as to increase the fitting ability of the network (each convolution layer is connected with BN layer and ReLU layer). After that, the feature images of $1 \times 1 \times 1$ were transformed into one-dimensional features by the global average pooling network. Finally, six regression values were

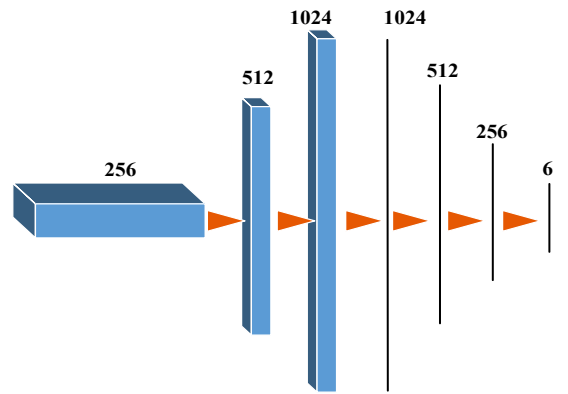


Figure 4. Architecture of the control point location network

output by three fully connection layers to locate two control points (X, Y and Z coordinates of two control points) on the navigation pedicle screw surgery path.

Data augmentation: For each input spinal image, we used image rotation, random cropping, flipping, random noise and other strategies for data augmentation. After data augmentation, multiple images with a size of $128 \times 128 \times 128$ are taken from each image and saved in a list. The images in the list are shuffled and packed into batches to feed into the network.

Training process: During the training process, mean square error (MSE) loss is adopted in this study as the loss function of network to fit the location between control points learned by the network and target points, and L2 normal form is adopted to increase the generalization ability of the network. The mean square error loss function is as follows:

$$J(w, b) = \frac{1}{2N} \sum_i |y - a|^2 \quad (2)$$

where $a = f(w \cdot x + b)$, x is the input of the network, w and b are the parameters of the network, and $f(\cdot)$ is the activation function. The loss function is optimized with the Adam solver, and the learning rate of the network is set to 0.0001, which enables the network to find a better solution out of the local optimal solution while rapidly converging.

Testing process: During the testing process, spinal images of $128 \times 128 \times 128$ are input into the neural network after the normalization. In the spinal pedicle screw surgery path planning, the network output has 6 values, the first 3 values are the coordinates of the surgical entry point, and the last 3 values are the coordinates of the surgical direction point.

III. EXPERIMENTS AND RESULTS

A. Experimental Design

In this study, 21 sets of spinal CT images were used, and the VB L1 of each spine contained two control points labeled by surgeons on the spinal pedicle screw surgical path. The original CT image and target image are resized to $128 \times 128 \times 128$ and normalized into 0-1 before put into network so as to ensure the stability of the network. We conducted a leave-three-out cross-validation study to evaluate the performance of the present method. More specifically, each time we randomly take 3 out of the 21 CT data as the testing data and the remaining 18 CT data as the training data. The process was repeated for 5 folds. In each fold, the segmented results and the located results of the testing data were compared with the associated ground truth segmentation and location.

The experimental procedure was divided into two stages: segmentation stage and location stage. In the segmentation stage, for each vertebra in a testing data of CT, we evaluated the volume overlap measurements using the Dice coefficient (DC) and the Jaccard coefficient (JC) between the surface models extracted from different segmentation. In the location stage, the mean square error (MSE) loss was used to evaluate the positioning results.

B. Experimental Environment

The proposed work was implemented in python using TensorFlow and trained on a desktop with a 3.6GHz Intel i7 CPU and a GTX 1080Ti graphics card with 11 GB GPU memory.

C. Experimental Results

For each validation experiment in five folds, the segmentation network can achieve successful segmentation from spinal CT images. Segmentation results of the cross validation study are shown in Table 1, where the results on each individual vertebra as well as on the entire lumbar region are presented. Our approach achieves a mean DC of $96.67 \pm 0.47\%$ on the entire lumbar region, which is an improvement compared with results obtained from original 3D U-Net that shown in Table 2, The location network can locate the control points in spinal pedicle screw surgery path and the margin of error is much lower than the margin of error allowed by surgery. Quantitative results of location of the cross validation study are shown in Table 2. A mean MSE of 1.34 voxels is achieved in the prediction of surgical path (Table 3). We also compared our results with the results of 3D Resnet101 shown in Table 4 to verify the effectiveness of our network. It can be seen that our network achieved similar results with much less computation than 3D Resnet101. Through testing, it is proved that the network can accurately locate the control points in the spinal pedicle screw surgical path, and reconstruct the complete pedicle screw surgical path according to the coordinates of control points.

TABLE I. SEGMENTATION RESULTS OF THE 5-FOLD CROSS VALIDATION ON SPINAL CT IMAGES USING OUR APPROACH

	L1	L2	L3	L4	L5	avg.
DC	97.03 ± 0.35	96.42 ± 0.56	96.15 ± 0.33	96.33 ± 0.68	96.86 ± 0.73	96.67 ± 0.47
JC	93.47 ± 0.82	92.98 ± 0.91	93.15 ± 0.62	92.76 ± 0.71	92.88 ± 0.54	93.05 ± 0.62

TABLE II. SEGMENTATION RESULTS OF THE 5-FOLD CROSS VALIDATION ON SPINAL CT IMAGES USING 3D-UNET

	L1	L2	L3	L4	L5	Avg.
DC	96.11 ± 0.45	95.33 ± 0.53	95.25 ± 0.37	95.41 ± 0.61	95.66 ± 0.72	95.55 ± 0.57
JC	92.17 ± 0.93	91.78 ± 0.95	92.35 ± 0.88	91.52 ± 0.77	91.78 ± 0.57	91.92 ± 0.71

TABLE III. LOCATION RESULTS OF THE 5-FOLD CROSS VALIDATION ON SPINAL CT IMAGES USING OUR APPROACH (IN VOXEL)

	1	2	3	4	5	avg.
training MSE	0.35	0.27	0.38	0.33	0.31	0.328
validation MSE	1.34	1.30	1.42	1.27	1.37	1.340

TABLE IV. LOCATION RESULTS OF THE 5-FOLD CROSS VALIDATION ON SPINAL CT IMAGES USING RESNET STRUCTURE (IN VOXEL)

	1	2	3	4	5	Avg.
training MSE	0.34	0.27	0.36	0.31	0.29	0.314
validation MSE	1.35	1.29	1.44	1.30	1.38	1.351

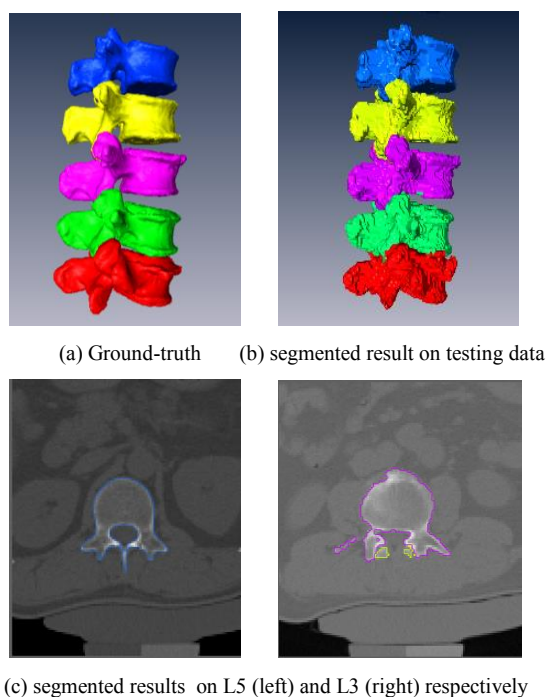


Figure 5. Automatic learning of spinal segmentation.

The segmentation results were shown in Fig. 5. Fig. 5(a) represents the target segmentation diagram and Fig. 5(b) represents the segmentation result generated by network. The five colors in this image represent five segments of the spine. As can be seen from the images, our segmentation network can well segment the spine in the CT images and separate the vertebral body from each other. The network is also able to segment the spinal column very well for the indistinguishable individuals, which has great advantages over the traditional segmentation method. Fig 5(c) shows the slices of different VBs on segmentation diagram generated by network. There is still some problems to be solved in the segmentation results: the generated diagrams are not smooth compared to the ground truth and need further processing.

Control point location results were shown in Fig. 6. As we can see, the location network can accurately locate the entry point and orientation point of spinal pedicle screw surgery, and obtain the spinal surgery path based on two points. The average error of surgical path generated by location network is much lower than that required by surgery, which proves the effectiveness and stability of our network.

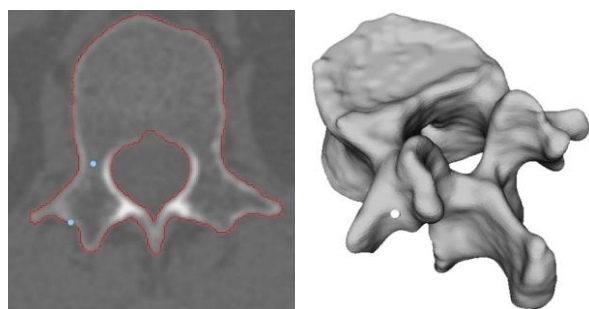


Figure 6. Automatic learning of surgical path points on spine.

Through the analysis of experimental results, we found that there is still a certain overfitting phenomenon in the network, which may be caused by the following reasons: The amount of CT data used for training is small and the data diversity is insufficient; there is room for improvement in our network structure. And these problems are also our next improvement direction.

In each fold, it took about 12 hours to finish the training for the segmentation network, and another 4 hours to finish the training of location network. After training, it took on average about 10 seconds to finish the segmentation of one testing data of CT imaging and 2 seconds to locate the control points of the surgery path.

IV. DISCUSSION AND CONCLUSIONS

We presented an automatic path planning method for spinal pedicle screw surgery and achieved acceptable results on 3D spinal CT images. Through learning, deep neural network-based method can automatically segment each VB of the spine and locate two control points of surgical path on the spine image data, which can facilitate personalized path planning for different surgical indications and the results are very close to the path manually planned by surgeons.

There are still several factors which influence the quality of the automatic planning compared to that of surgeons' manual planning. (1) Due to the limitations of machines, the segmentation network and location network we designed are small consider to other segment network and the size of the input spinal image is $128 \times 128 \times 128$, which is not large enough and should be further optimized. (2) For the mask labelling of training set, there are differences among different surgeons and surgical paths defined by surgeons in different states, which will affect stability in network training. (3) Different spinal images always have different resolutions which will affect the learning results. Therefore, the follow-up research should add more quantitative information in the training set and conduct data cleaning to ensure the accuracy of path planning.

The network depth in the proposed scheme is not deep enough and should be further increased. For the segmentation network, residual block should be added into the network to enhance the learning ability of the network. At the same time, other loss functions such as the focal loss can be used to deal with the imbalance problems between background voxels and vertebra voxels. For the location network, a more complicated and advanced structure should be used to extract features on 3D spinal images in further research, which allows for more accurate location of key points in spinal surgical path.

In our research, only *point-based* surgical path planning for navigated pedicle screw surgery is achieved using the deep neural network. Considering that the form of surgical planning result on the spinal CT image is not only a line, but also a region, or a volume, path planning of spinal surgery based on *region* or *volume* should also be considered, which can improve the generalization ability of our method.

In the automatic planning process of spinal surgery, we only performed anatomic image analysis without considering other

factors such as the biomechanical influence of the planned path. Considering the rationality of surgical path from perspective of biomechanics, although surgeons have implicitly considered biomechanical factors in the manual planning of surgical path, whether the path learned by deep neural network can achieve the optimal biomechanical distribution needs to be further studied and validated.

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